
Endomorphisms of Hopf Algebras and a Little Bit of Control

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Summary. Firstly this paper summarizes some of the relations between control theory and coalgebra and Hopf algebra theory, particularly the fact that the cofree coalgebra over a finite rank free module consists precisely of the realizable power series in noncommuting variables. The second half of the paper concerns generalizations of the Hopf algebra of permutations which in turn generalizes the Hopf algebras of noncommutative symmetric functions and the Hopf algebra of quasisymmetric functions.

Dedication. To Clyde F Martin on the occasion of his 60-th birthday and to the world of control which somehow manages to sneak in almost everywhere.

8.1 Introduction

It is relatively well known that system and control theory, mostly the part that is known as realization theory, and Hopf algebras have something to do with one another. Two independent standard references in this direction are [13, 15]. See also [14].

As a matter of fact the cofree coalgebra over a module perfectly captures on the one hand the idea of recursiveness for power series (in several noncommuting variables) and also the idea of realizability. For a precise statement see section 2 below.

This, however, is not the real subject of this talk and merely serves to point out that there are fundamental links between the worlds of Hopf algebras and that of control. A number of other relevant references here are [2, 3, 7, 10, 11, 12, 14, 19]. The most important Hopf algebra in all these papers is *Shuffle*, the Hopf algebra of all words in a given alphabet as basis, the shuffle product as multiplication and cut as comultiplication. *Shuffle* will be described in detail below as it is also a basic ingredient in the construction of the Hopf algebras of endomorphisms which form the main topic of this talk.

The real aim of this talk is to describe a number of very beautiful Hopf algebras in the fond and not unreasonable hope that one day they will find applications in systems and control and elsewhere.

Before that I take time out to discuss the notion of multivariable recursiveness and its relations with control (realization theory type) ideas on the one hand and free coalgebras on the other hand.

8.2 Recursiveness

Let $\{X_a : a \in I\}$ be a finite or infinite set of indeterminates indexed by a set I . Let I^* denote the monoid (under concatenation) of all words over the set I (including the empty word (which serves as the unit element)). An element of I^* is usually written in these pages as

$$\alpha = [a_1, \dots, a_n] \quad \text{where } a_i \in I$$

The length of such a word is m , $\lg(\alpha) = m$. A power series in the X_a , $a \in I$ over a commutative ring with unit element k is simply a formal expression

$$f = \sum_{\alpha \in I^*} f_\alpha X_\alpha \quad (8.1)$$

where the coefficients f_α are in k and X_α is short for $X_{a_1}, X_{a_2}, \dots, X_{a_n}$. The degree n component of f is

$$f^n = \sum_{\lg(\alpha)=n} f_\alpha X_\alpha \quad (8.2)$$

Equivalently, a power series is a function $I^* \rightarrow k$, $\alpha \mapsto f(\alpha) = f_\alpha$.

In the case of one variable, everyone knows it means for a power series to be recursive. There are obvious generalization to the cases of more variables. For instance:

f is left polynomially recursive if there is a finite set of monomials X_{λ_j} , $j = 1, \dots, l$, with corresponding coefficients c_j such that for each large enough n

$$f^n = \sum_{j=1}^l c_j X_{\lambda_j} f^{n-\lg(\lambda_j)} \quad (8.3)$$

and there is similar notion of right polynomial recursiveness. These notions, however, turn out not to be general enough.

A power series f over k , see (8.1), is left recursive if there is a finite set of monomials X_{λ_i} , $i = 1, \dots, l$ and for some fixed $s > \max\{\lg(\lambda_i), i = 1, \dots, l\}$, there are coefficients $c_{\gamma,i} \in k$, for each $i \in \{1, \dots, l\}$ and word $\gamma \in I^*$ of length s , such that for $n \geq s$ for each $\alpha \in I^*$ of length n

$$f(\alpha) = \sum_{i=1}^l c_{\alpha \text{ pre}(s), i} f(\lambda_i \alpha_{\text{suf}}) \quad (8.4)$$

where if β, γ are two words over J , $\beta\gamma$ is the concatenation of them and where for a word α of length $\geq s$, $\alpha_{\text{pre}(s)}$ is the prefix of α of length s and α_{suf} is the corresponding suffix (or tail), so that $\alpha = \alpha_{\text{pre}(s)}\alpha_{\text{suf}}$.

There is of course an entirely analogous notion of right recursive. Note that these notions of recursiveness exactly capture the idea of recursiveness in the sense that a coefficient $f(\alpha)$ for $\text{lg}(\alpha)$ large enough is (from the left or the right) a linear combination of coefficients for words of lesser length in a uniform manner (same coefficients).

A power series (8.1) is left Schützenberger recursive if and only if there is a finite nonempty set of words S closed under taking prefixes (so that in any case the empty word is in S) such that for all words $\beta \in T = SX \setminus S$ there are coefficients $a_{\beta\alpha}$, $\alpha \in S$, $\beta \in T$ such that for all words γ

$$f_{\beta\gamma} = \sum_{\alpha \in S} a_{\beta\alpha} f_{\alpha\gamma} \quad (8.5)$$

At first sight this does not look all that recursive. For one thing the word $\alpha\gamma$ occurring on the right hand side of (8.5) may very well have longer length than the word $\beta\gamma$. However, for each word ω let α be the longest prefix of ω that is in S , and write $\omega = \alpha\omega'$, then (8.5) is recursive with respect to the length of ω' . Again, there is a corresponding notion of right Schützenberger recursiveness.

Given a power series (8.1) consider the infinity by infinity matrix with rows and columns indexed by the monoid of words I^* whose entry at location (α, β) is the coefficient $f_{\alpha\beta}$. This matrix is called the Hankel matrix of f .

Let $A\langle X_\alpha : \alpha \in I \rangle$ be the free associative algebra over k in the indeterminates X_α . The monomials X_α , $\alpha \in I^*$ form a basis (as a k -module) for $k\langle X_\alpha : \alpha \in I \rangle$. A power series f as in (8.1) is the same thing as a k -module morphism $k\langle X_\alpha : \alpha \in I \rangle \rightarrow k$, viz the one that assigns to X_α the coefficient f_α . A power series f is called representative if there is a finite number n and power series g_i, h_i , $i = 1, \dots, n$ such that for each $\alpha, \beta \in I^*$

$$f_{\alpha\beta} = \sum_{i=1}^n g_i \alpha h_i \beta \quad (8.6)$$

The origin of the terminology is obvious. If ρ is a representation of the monoid I^* in the $n \times n$ matrices over k then the entries of ρ as functions on I^* have property (8.6).

Finally, to end this string of definitions, the power series f is called realizable, called recognizable in the automata theory world, if there is a number N and $N \times N$ matrices $\rho(X_a)$, $a \in I$, an $N \times 1$ matrix b (column vector), an $1 \times N$ matrix c (row vector) such that for all words $\alpha = [a_1, \dots, a_m]$

$$f_{\alpha} = c\rho(X_{a_1}) \cdots \rho(X_{a_n})b \quad (8.7)$$

A fundamental theorem is now:

Theorem 1. *Let I be finite and let k be a Noetherian integral domain. Then the following seven properties for a power series f are equivalent*

- (i) f is left recursive
- (ii) f is right recursive
- (iii) f is left Schützenberger recursive
- (iv) f is right Schützenberger recursive
- (v) $H(f)$ has finite rank
- (vi) f is representative
- (vii) f is realizable.

The proof is a mixture of ideas and constructions of Schützenberger, see [5, 22], Fliess, [9], and Rouchaleou, see [20]. See [16] for details.

Let $FPS(\mathfrak{X})$ be the k -module of all formal power series in the set of indeterminates $\mathfrak{X} = \{X_a : a \in I\}$. The polynomials in $FPS(\mathfrak{X})$ are those power series for which only finitely many coefficients are nonzero¹.

By definition the submodule of rational power series, $FPS_{rat}(\mathfrak{X})$, is the smallest submodule that contains the polynomials and is closed under multiplication and taking inverses when possible (which is exactly when the constant term is an invertible element of k). The famous Kleene-Schützenberger theorem now says:

Theorem 2. *A power series in a finite number of variables over a Noetherian integral domain is rational if and only if it is realizable.*

Again, see [16] for a proof.

Now consider the following decomposition structure

$$X_{\alpha} \mapsto \mu(X_{\alpha}) = \sum_{\beta\gamma=\alpha} X_{\beta} \otimes X_{\gamma}$$

or more precisely

$$\mu(f) = \sum_{\alpha, \beta\gamma=\alpha} f_{\alpha} X_{\beta} \otimes X_{\gamma} \quad (8.8)$$

As a rule it is not possible to write $\mu(f)$ as a *finite* $\sum_j g_j \otimes h_j$ but it can happen; for instance this always happens for polynomials. As a matter of fact, practically by definition, $\mu(f)$ can be written as such a finite sum iff f

¹ Warning. In case there is an infinity of indeterminates there is another possible definition of polynomials, namely those power series f for which its n -th degree component is zero for n large enough. This one is also appropriate in some contexts and is for instance used when discussing symmetric functions (in an infinity of indeterminates, which is the proper context for symmetric functions).

is representative. And then the equivalence “representative” $\Leftrightarrow$ “realizable” immediately gives that $\mu(f)$ can be written as a finite sum (with possibly different g_j, h_j)

$$\mu(f) = \sum_j g_j \otimes h_j, \quad g_j, h_j \in FPS_{\text{rat}}(\mathcal{X})$$

(where theorems 2 and 1 are used to identify $FPS_{\text{rat}}(\mathcal{X}) = FPS_{\text{representative}}(\mathcal{X}) = FPS_{\text{realizable}}(\mathcal{X})$). Thus the decomposition structure induces a morphism of modules (a comultiplication)

$$\mu : FPS_{\text{rat}}(\mathcal{X}) \rightarrow FPS_{\text{rat}}(\mathcal{X}) \otimes FPS_{\text{rat}}(\mathcal{X}) \quad (8.9)$$

There is also an augmentation or counit

$$\varepsilon : FPS_{\text{rat}}(\mathcal{X}) \rightarrow k, \quad f \mapsto f^0 \quad (8.10)$$

and (8.9), (8.10) combine to define a coalgebra structure on $FPS_{\text{rat}}(\mathcal{X})$ in the sense of the following definition.

Definition 1. A coalgebra over k is a k -module C together with two k -module morphisms $\mu : C \rightarrow C \otimes_k C$ (called comultiplication) and $\varepsilon : C \rightarrow k$ (called counit) such that

$$\begin{aligned} (\text{id} \otimes \mu)\mu &= (\mu \otimes \text{id})\mu \text{ (coassociativity)} \\ (\varepsilon \otimes \text{id})\mu &= \text{id}, (\text{id} \otimes \varepsilon)\mu = \text{id} \text{ (counit property)} \end{aligned} \quad (8.11)$$

where all tensor products are over k and the canonical identifications $C \otimes_k k = C = k \otimes_k C$ are used.

This notion is dual to that of an algebra as is seen by writing out the definition of an algebra in diagram form and reversing all arrows. See any standard text on Hopf algebras, e.g. [1, 8, 18, 23].

The coalgebra $FPS_{\text{rat}}(\mathcal{X})$ comes with a natural morphism of k -modules

$$\pi : FPS_{\text{rat}}(\mathcal{X}) \rightarrow \prod_{a \in I} kX_a, \quad f \mapsto f^1$$

making it the cofree coalgebra over the module $\prod_{a \in I} kX_a$ in the sense of the following universal property²

For every coalgebra C over k and morphism of k -modules $\varphi : C \rightarrow \prod_{a \in I} kX_a$ there is a unique morphism of coalgebras $\tilde{\varphi} : C \rightarrow FPS_{\text{rat}}(\mathcal{X})$ such that $\pi\tilde{\varphi} = \varphi$.

Thus $FPS_{\text{rat}}(\mathcal{X})$ is a kind of universal receptor for coalgebra structures, which goes some way to explaining its dominant role in control and automata theory.

²When I is finite this module is simply the free module on the finite basis formed by the $X\tilde{O}$ s. When I is infinite this is not the case.

For k a field and M the free module over k with basis $\{X_a : a \in I\}$ the theorem that $\hat{T}M_{\text{repr}}$ is the cofree coalgebra over M is known as the Block-Leroux theorem, [6]. Here, if I is finite, $\hat{T}M_{\text{repr}} = FPS_{\text{repr}}(\mathfrak{X})$; but if I is infinite $\hat{T}M$ is a submodule (much smaller) of $FPS_{\text{repr}}(\mathfrak{X})$.

There are several multiplication structures on $FPS_{\text{rat}}(\mathfrak{X})$ that make it into a Hopf algebra (see below for this notion). For instance the shuffle product and the Hadamard product. For control the shuffle product seems to be the most relevant.³

For an infinity of indeterminates there are various analogues of theorems 1 and 2. All kinds of finiteness conditions come into play, not only vis à vis the definitions of recursive but also with respect to what kind of power series f are considered. See [16].

8.3 The Idea of a Hopf Algebra

Before stating the formal definition, let me try my hand at giving some sort of informal and intuitive idea of what a Hopf algebra is.

Intuitively:

Algebra = something with a composition.

Coalgebra = something with a decomposition.

Hopf algebra = something with both a composition and a decomposition in a compatible way.

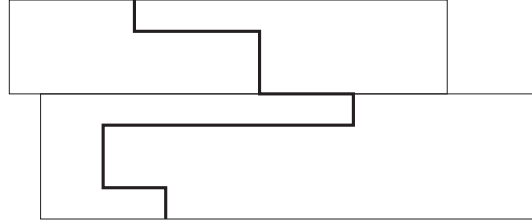
The most important compatibility condition is:

Comultiplication respects multiplications (and vice versa).

This is often called the Hopf property.

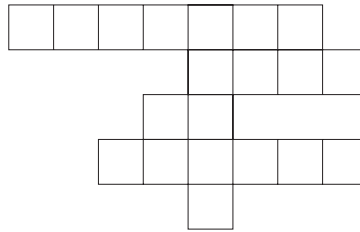
It does not say that the comultiplication undoes the multiplication (or vice versa) but rather that they are sort of orthogonal to each other. Think of the comultiplication as giving all ways of composing an object into a left part and a right part and of multiplication as vertical composition. Then, in words, the condition says that all left parts of the decomposition of a product are all products of left parts of the two factors and the corresponding right parts are the products of the corresponding right parts of the decompositions of the factors involved. In pictures (stacking things vertically is composition; cutting things into left and right parts is decomposition):

³The results reported on in this section shed additional light on the relations between control theory and Hopf algebras. It is certainly significant that the realizable power series precisely form a cofree coalgebra with respect to the alphabeth concerned. This does not mean that there is now a nice coherent well organized picture of all these matters. In fact, organizing what is known from the references cited, and filling in what is missing, is a nice potential PhD level research topic.



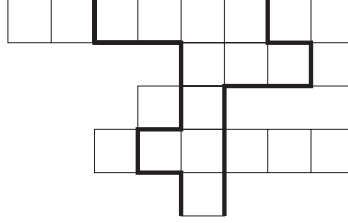
Here the top box is object 1 and the bottom one is object 2. The two boxes together are the composition of object 1 and object 2 (in that order). The first three segments of the jagged heavy line give a decomposition of object 1 into a left part L1 and a right part R1 and the last five segments of that heavy jagged line give a decomposition into a left part L2 and a right part R2 of object 2. The complete heavy broken line gives the corresponding decomposition of the product of object 1 and 2 into a left part that is the product of L1 and L2 and a right part that is the product of R1 and R2.

In fact there is a most important and beautiful bialgebra (which is in fact a Hopf algebra) that looks almost exactly like this. Consider stacks of rows of unit boxes.



Here two such stacks are considered equivalent if they have the same number of boxes in each layer. Thus it only matters how many boxes there are in each layer. Empty layers are ignored. The case depicted is hence given by the word $[7, 4, 2, 6, 1]$ over the positive integers N . The empty word is permitted and corresponds to any stack of empty layers. The possible decompositions of a stack are obtained by cutting each layer into two parts. Two of these, for the example at hand, are indicated below.

These two correspond to the decompositions $[2, 1, 1] \otimes [5, 4, 1, 5, 1]$ and $[6, 3, 2, 3, 1] \otimes [1, 1, 3]$. A convenient way of encoding this algebraically is to consider the free associative algebra $\mathbf{Z}(Z)$ over the integers in the indeterminates $Z_1, Z_2, Z_3, \dots$. A basis (as a free Abelian group) for this algebra is given by 1 and all (noncommutative) monomials $Z_{i_1}, Z_{i_2}, \dots, Z_{i_m}$. This monomial encodes the stack with layers of $i_1, i_2, \dots, i_m$ boxes. Thus the example above corresponds to the monomial $Z_7 Z_4 Z_2 Z_6 Z_1$. The comultiplication on $\mathbf{Z}(Z)$ is given by the algebra homomorphism determined by



$$\mu(Z_n) = \sum_{i+j=n} Z_i \otimes Z_j \text{ where } Z_0 = 1 \text{ and } i, j \in \mathbf{N} \cup \{0\} = \{0, 1, 2, 3, \dots\} \quad (8.12)$$

This, together with the counit

$$\varepsilon(Z_n) = 0, n \geq 1 \quad (8.13)$$

defines the bialgebra (Hopf algebra) $NSymm$ of noncommutative symmetric functions.

To conclude this section, here are the formal definitions of bialgebras and Hopf algebras.

Definition 2 (Bialgebra). A bialgebra over k is a k -module H equipped with four module morphisms $m : H \otimes H \rightarrow H$ (multiplication, composition), $e : k \rightarrow H$ (unit element), $\mu : H \rightarrow H \otimes H$ (comultiplication, decomposition), $\varepsilon : H \rightarrow k$ (counit) such that

- (i) (H, m, e) is an (associative) algebra (with unit element $e(1)$)
- (ii) (H, μ, ε) is a (coassociative) coalgebra (with counit, see (8.11) above)
- (iii) μ and ε are algebra morphisms

or, equivalently (iii)' m and e are coalgebra morphisms.

Definition 3 (Hopf algebra). A Hopf algebra over k is a bialgebra H with one extra morphism $\iota : H \rightarrow H$, called antipode, such that

$$m(\iota \otimes \text{id})\mu = e \text{boldsymbol{\varepsilon}}, m(\text{id} \otimes \iota)\mu = e \boldsymbol{\varepsilon}$$

The morphism μ is something very much like the inverse in group theory.

8.4 The Shuffle Hopf Algebra

Consider the monoid N^* all words over the alphabeth $\mathbb{N}$ of positive integers (including the empty word) and let *Shuffle* be the free $\mathbf{Z}$ -module (Abelian group) with basis N^* . The shuffle product of two words $\alpha = [a_1, a_2, \dots, a_m]$ and $\beta = [b_1, b_2, \dots, b_n]$ is defined as follows. Take a sofar empty word with $n+m$ slots. Choose m of the available $n+m$ slots and place in it the natural numbers from α in their original order; place the entries from β in their original order

in the remaining n slots. The product of the two words α and β is the sum (with multiplicities) of all words that can be so obtained. So, for instance

$$\begin{aligned} [a, b] \times_{\text{sh}} [c, d] &= [a, b, c, d] + [a, c, b, d] + [a, c, d, b] \\ &\quad + [c, a, b, d] + [c, a, d, b] + [c, d, a, b] \\ [1] \times_{\text{sh}} [1, 1, 1] &= 4[1, 1, 1, 1] \end{aligned}$$

This defines a commutative associative multiplication on *Shuffle* for which the empty word is a unit element. Moreover with cut as a comultiplication

$$\mu([a_1, a_2, \dots, a_m]) = \sum_{i=0}^m [a_1, \dots, a_i] \otimes [a_{i+1}, \dots, a_m]$$

count $\varepsilon([\])=1$, $\varepsilon(\alpha)=0$ if $\lg(\alpha) \geq 1$, where the length of a word $\alpha = [a_1, \dots, a_m]$ is $\lg(\alpha) = m$, and antipode

$$\iota([a_1, a_2, \dots, a_m]) = (-1)^m [a_m, a_{m-1}, \dots, a_1]$$

Shuffle becomes a Hopf algebra. It is not difficult to verify this directly. Some remarks re *Shuffle*

- Over a field of characteristic zero *Shuffle* is free. Over the integers this is definitely not the case
- There is a natural quantum version, *qShuffle* which interpolates between concatenation and *Shuffle* as regards its algebra structure.

As to both remarks I do not know what they could imply for control.

8.5 The Hopf Algebra of Permutations of Reutenauer, Malvenuto, Poirier (the MPR Hopf Algebra)

As an Abelian group MPR it is the free Abelian group with as basis all permutations on n letters, all n

$$\begin{aligned} MPR &= \oplus_{n=0}^{\infty} \mathbf{Z}S_n, \\ \mathbf{Z}S_0 &= \mathbf{Z}, \mathbf{Z}S_1 = \mathbf{Z}[1], \mathbf{Z}S_2 = \mathbf{Z}[1, 2] \oplus \mathbf{Z}[2, 1], \dots \end{aligned} \tag{8.14}$$

Permutations are written as special words

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & m \\ a_1 & a_2 & \cdots & a_m \end{pmatrix} = [a_1, a_2, \dots, a_m]$$

Thus a word over the natural numbers is a permutation if and only if it is injective (no letter occurs more than once) and surjective (there are no gaps, meaning that if $\text{supp}(\alpha) = \{a_1, \dots, a_m\}$ is the support of α then $\text{supp}(\alpha)$ is an initial interval $\{1, \dots, m'\}$ for some $m' \in \mathbf{N}$).

Multiplication example

$$[2, 1] \times_{MPR} [1, 2] = [2, 1, 3, 4] + [2, 3, 1, 4] + [2, 3, 4, 1] + [3, 2, 1, 4] + [3, 2, 4, 1] + [3, 4, 2, 1]$$

Comultiplication example

$$\mu_{MPR}([2, 4, 3, 1]) = [] \otimes [2, 4, 3, 1] + [1] \otimes [3, 2, 1] + [1, 2] \otimes [2, 1] + [1, 3, 2] \otimes [1] + [2, 4, 3, 1] \otimes []$$

The rules behind these examples are as follows. Given a word $\alpha = [a_1, a_2, \dots, a_m]$ without repeated letters let $\text{st}(\alpha) = [b_1, \dots, b_m]$ be the permutation word on m letters defined by $b_i < b_j \Leftrightarrow a_i < a_j$. For instance $\text{st}([6, 8, 3, 5]) = [3, 4, 1, 2]$. Then the comultiplication on MPR is given by

$$\mu_{MPR}([a_1, \dots, a_m]) = \sum_{j=1}^m \text{st}([a_1, \dots, a_j]) \otimes \text{st}([a_{j+1}, \dots, a_m]) \quad (8.15)$$

The multiplication is easier to describe

$$m_{MPR}([a_1, \dots, a_m] \otimes [b_1, \dots, b_n]) = [a_1, \dots, a_m] \times_{sh} [m + b_1, \dots, m + b_n] \quad (8.16)$$

There is also an antipode. Using the grading, see (8.14), it is easy to prove that an antipode exists; writing down an explicit formula is a different matter, but see [4].

Once the formulas have been guessed it is not really difficult to prove that MPR is indeed a Hopf algebra. The question is more, where do these formulas come from and are there natural generalizations. The next sections give answers to these questions.

8.6 The Word Hopf Algebra, WHA

As an Abelian group WHA is the free Abelian group on all words over the natural numbers (including the empty word (which serves as the identity)). An example of the multiplication formula is

$$\begin{aligned} [2, 3, 2] \times_{WHA} [4, 1] &= [2, 3, 2, 7, 4] + [2, 3, 7, 2, 4] + [2, 7, 3, 2, 4] + [7, 2, 3, 2, 4] \\ &\quad + [2, 3, 7, 4, 2] + [2, 7, 3, 4, 2] + [7, 2, 3, 4, 2] + [2, 7, 4, 3, 2] \\ &\quad + [7, 2, 4, 3, 2] + [7, 4, 2, 3, 2] \\ &= [2, 3, 2] \times_{sh} [7, 4] \end{aligned}$$

So the multiplication is again a shifted shuffle product with the shift equal to $ht(\alpha) = \max\{a_1, \dots, a_m\}$. The comultiplication is more mysterious. Here is an example

$$\begin{aligned}\mu_{WHA}([3, 3, 2, 2, 2, 5, 9, 6]) = & [\] \otimes [3, 3, 2, 2, 2, 5, 9, 6] + [1, 1] \otimes [2, 2, 2, 4, 8, 5] + \\ & + [3, 3, 2, 2, 2] \otimes [2, 6, 3] + [3, 3, 2, 2, 2, 5] \otimes [4, 1] \\ & + [3, 3, 2, 2, 2, 5, 8] \otimes [1] + [3, 3, 2, 2, 2, 5, 9, 6] \otimes [\]\end{aligned}$$

and I do not think it is easy to guess what could be the general recipe behind this formula.

8.7 Convolution

The multiplication parts of *MPR* and *WHA* are relatively easy to understand. Let H be a Hopf algebra and let φ, ψ , be two endomorphisms of H (as Abelian groups, or vector spaces, of k -modules, as the case may be). Then a new endomorphism is obtained by convolution⁴ as follows

$$H \xrightarrow{\mu} H \otimes H \xrightarrow{\varphi \otimes \psi} H \otimes H \xrightarrow{m} H \quad (8.17)$$

Now take $H = \text{Shuffle}$ and interpret a permutation $\sigma \in S_m$ as the endomorphism of *Shuffle* that takes any word $\alpha = [a_1, \dots, a_m]$ of length m into $[a_{\sigma(1)}, \dots, a_{\sigma(m)}]$ and is zero on all other words. Thus, if $[s_1, \dots, s_m]$ is the word describing the permutation a word $\alpha = [a_1, \dots, a_m]$ gets taken into $[a_{s_1}, \dots, a_{s_m}]$. Then it is easy to see that convolution exactly gives the multiplication on *MPR*.

This also works for *WHA* for several different interpretations of words as endomorphisms of *Shuffle*.

For instance there is the very natural the interpretation that says that a word $\alpha = [a_1, \dots, a_m]$ acts on words from *Shuffle* as follows.

Let $ht(\alpha) = \max\{a_i\}$. Then α acts on words of length unequal to $ht(\alpha)$ as zero and a word $[u_1, \dots, u_{ht(\alpha)}]$ is taken to $[u_{a_1}, u_{a_2}, \dots, u_{a_m}]$.

There are other interpretations that one can think of, but for the multiplication this one works, as do some others (meaning that they induce the same multiplications. As it will turn out the interpretation just given for letting arbitrary words act as endomorphisms of *Shuffle* is not the right one.

8.8 Coconvolution

Everything in the Hopf algebra world tends to have something like a dual version. So let's try coconvolution. Given an endomorphism φ of a Hopf algebra H look at

$$H \otimes H \xrightarrow{m} H \xrightarrow{\varphi} H \xrightarrow{\mu} H \otimes H$$

⁴In the case of certain Hopf algebras coming from groups this can indeed be seen to be classical convolution, see [18].

This give an element of $End(H \otimes H)$ which (for infinite dimensional things like *Shuffle*) is quite a different (much larger) object than $End(H) \otimes End(H)$.⁵ Still there might be a suitable projection or something. Let's try an example

$$\begin{aligned} [u1, u2] \times_{sh} [v1, v2, v3] &= [u1, u2, v1, v2, v3] + [u1, v1, u2, v2, v3] \\ &+ \cdots + [v1, v2, v3, u1, u2] \end{aligned}$$

which under the permutation $\sigma = [3, 1, 4, 5, 2]$ goes to

$$[v1, u1, v2, v3, u2] + [u2, u1, v2, v3, v1] + \cdots + [v3, v1, u1, u2, v2]$$

Apply cut to this and look for terms with support $\{u1, u2\} \otimes \{v1, v2, v3\}$. There is just one such, viz

$$[u2, u1] \otimes [v2, v3, v1] = [2, 1] \otimes [2, 3, 1]([u1, u2] \otimes [v1, v2, v3])$$

which fits with the comultiplication on *MPR*. Indeed the (length = 2) tensor (length = 3) component of $\mu([3, 1, 4, 5, 2])$ is

$$st([3, 1]) \otimes st([4, 5, 2]) = [2, 1] \otimes [2, 3, 1]$$

And, indeed, quite generally, this procedure produces formula (8.15). It is, however, a quite dodgy procedure. There is no guarantee whatever that the resulting comultiplication will indeed yield a Hopf algebra. In this case it does. In other cases it does not, as in the example discussed in the next section.

8.9 Generalization to Words (Not Just Permutations)?

There is a perfectly nice way, already used, to make any word, not just permutations, act on *Shuffle*.

Let $ht(\alpha) = \max\{a_i\}$, $\alpha = [a_1, a_2, \dots, a_m]$. Then acts on words of the form $[u_1, \dots, u_{ht(\alpha)}]$ by taking such a word to $[u_{a_1}, \dots, u_{a_m}]$.

Does this also give a nice Hopf algebra. Perhaps *WHA* from above.

Answer: NO, the resulting comultiplication does not satisfy the Hopf property with respect to convolution.

⁵When H is a free finite rank module over k , $End(H) = H \otimes H^*$ and $End(H \otimes H) = End(H) \otimes End(H)$. Then coconvolution defines a comultiplication on $End(H) = H \otimes H^*$. Moreover, convolution and coconvolution combine to define a Hopf algebra structure on $End(H)$. The resulting Hopf algebra is in fact the tensor product of the Hopf algebra H and its dual H^* .

This Hopf algebra comes with a second multiplication (composition of endomorphisms) and a second comultiplication (cocomposition of morphisms). There are also a unit and counit for this, viz

evaluation : $H \otimes H^* \rightarrow k$ and its dual coevaluation. The second multiplication (comultiplication) may or may not be distributive over the first one. But in any case these special Hopf algebras have a richer structure than the standard ones. As far as I know nothing much has been done in the way of systematic investigation of these Hopf algebras of endomorphisms.

8.10 The Right Generalization to Words

Here is a sufficiently complicated example

$$[9, 10, 9, 5, 3, 13, 12, 3] \leftrightarrow \begin{pmatrix} [u_1, u_1, u_1, u_2, u_2, u_3, u_3, u_3, u_3, u_4, u_5, u_5, u_6] \\ [u_3, u_4, u_3, u_2, u_1, u_6, u_5, u_1] \end{pmatrix} \quad (8.18)$$

This works as follows. First look at the alphabeth of letters occurring in the left hand side. The smallest integer occurring is 3. This skips 1,2 and together with 3 accounts for 3 u_1 's; the next one is 5, skipping 4, given rise to 2 u_2 's; the next one is 9, skipping 6,7,8, giving rise to 4 u_3 's; then comes 10, nothing skipped so 1 u_4 ; then 12, skipping 11, so 2 u_5 's; and finally 13, nothing skipped, so 1 u_6 . This describes the first row of the right hand side of the correspondence. The second row is obtained by writing down the 9-th, 10-th, 9-th, 5-th, 3-rd, 13-th, 12-th, 3-rd item from the top row.

It is clear how to go from the right hand side of (8.18) to the left hand side; the top row determines the support of the left hand side and then the bottom row determines the word itself.

The meaning is that a word like the left hand side acts on *Shuffle* as zero on all basis elements except those of the form like the right hand side of (8.18) (for the case at hand) and acts as indicated on the right hand side on basis elements of the shape indicated.

It is somewhat important that we are really dealing with recipes for endomorphisms rather than with concrete endomorphisms. This means that a word acts on all words of one specified shape in a uniform way.

Now apply the same dodgy procedure of section 8 above and the result is a Hopf algebra WHA . Proving that it is actually a Hopf algebra is a bit of a mess though. Fortunately there is a further generalization for which the proof is much easier and which contains this one as a sub Hopf algebra.

8.11 The Double Word Hopf Algebra, $2WHA$

The most general, so far, of these Hopf algebras is the double word Hopf algebra, $2WHA$. It is defined as follows (over the integers or any other base ring). A basis for the underlying free module is formed by pairs of words in indeterminates $x_1, x_2, \dots$

$$q = \begin{pmatrix} \gamma \\ \gamma' \end{pmatrix}, \quad \text{supp}(\gamma) = \text{supp}(\gamma') \quad (8.19)$$

These are symbolic substitutions, not elements of e.g. *Shuffle*. Only the pattern matters, not the actual symbols used. For instance

$$\begin{pmatrix} [x_1, x_1, x_2] \\ [x_1, x_2, x_1] \end{pmatrix} = \begin{pmatrix} [y_1, y_1, z_2] \\ [y_1, z_2, y_1] \end{pmatrix}$$

The multiplication is defined as follows: Let

$$p = \begin{pmatrix} \beta \\ \beta' \end{pmatrix}, \quad \text{supp}(\beta) = \text{supp}(\beta')$$

and

$$q = \begin{pmatrix} \gamma \\ \gamma' \end{pmatrix}, \quad \text{supp}(\gamma) = \text{supp}(\gamma')$$

be two symbolic word transformations (recipes for transforming words). First, if necessary rewrite them so that there are no common symbols between the two:

$$\text{supp}(\beta) \cap \text{supp}(\gamma) = \emptyset.$$

Then the product is

$$m(p \otimes q) = \begin{pmatrix} \beta * \gamma \\ \beta' \times_{sh} \gamma' \end{pmatrix} \quad (8.20)$$

Now let's define the comultiplication: A good cut of a symbolic word β is a cut $\beta = \beta' * \beta''$ such that

$$\text{supp}(\beta') \cap \text{supp}(\beta'') = \emptyset$$

The comultiplication is now

$$\mu(q) = \sum_{\substack{\text{good cuts,} \\ \gamma' * \gamma'' = \gamma}} \begin{pmatrix} q^{-1}(\gamma'') \\ \gamma'' \end{pmatrix} \otimes \begin{pmatrix} q^{-1}(\gamma''') \\ \gamma''' \end{pmatrix} \quad (8.21)$$

Here $q^{-1}(\gamma'')$ is that subword of γ that is obtained by removing all letters that are not in γ'' . This is a Hopf algebra⁶ that vastly generalizes *MPR* and it has a lot of similar extra structure (and much more besides). First of all there are obvious inclusions

$$PMR \subset WHA \subset 2WHA$$

There is a (nondefinite) nondegenerate inner product:

$$\left(\begin{pmatrix} \beta \\ \beta' \end{pmatrix}, \begin{pmatrix} \gamma \\ \gamma' \end{pmatrix} \right) = \delta_{\beta\gamma'} \delta_{\beta'\gamma} \quad (8.22)$$

The Hopf algebra $2WHA$ is autodual with respect to this inner product, and so is its sub Hopf algebra *MPR*.

There are a second multiplication (composition of endomorphisms) and a second comultiplication (cocomposition of endomorphisms) with unit and counit. There is room for all kinds of endomorphisms and automorphisms

⁶The proof is straightforward and that immediately also gives proofs that *MPR* and *WHA* are Hopf algebras.

(Frobenius-like, Verschiebung-like). Unlike in the case of *MPR*. There quite a few natural sub Hopf algebras. For instance the ones composed of surjective pairs of words, of injective pairs of words, or the one for which the supports of γ and γ' are equal including multiplicities, or combinations of these restrictions. And there are a large number of natural Hopf algebra morphisms between all these; both inclusions and projections coming from various standardization maps, see [21]. All this comes from looking at endomorphism recipes for *Shuffle* and its dual Hopf algebra, *Lie Hopf*. There are other Hopf algebras, such as the dual pair *NSymm*, *QSymm*, for which this endomorphism recipe idea can be carried out (including *2WHA* itself). This remains totally uninvestigated at this time.

See [17] for much more detail on Hopf algebras of endomorphisms of Hopf algebras and proofs.

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